

AGENTCALL AI

Comprehensive Platform, Architecture & Scale Audit Report

Audit Date: September 2, 2026	Target Capacity: 100k+ Active Businesses
Stack: Next.js 14 / NestJS 10 / Postgres 18 / Redis 3	Status: Live & Connected Locally
Telephony Providers: Twilio / Exotel	AI Speech Pipeline: Deepgram + GPT-4o + ElevenLabs

1. Executive Summary

AgentCall AI is an enterprise Autonomous Voice Calling Agent OS designed to replace and scale outbound sales, lead qualification, inbound front-desk reception, appointment booking, and polite collections. Built on a multi-tenant microservices architecture, the application orchestrates a sub-800ms conversational audio loop using streaming WebSockets, neural speech recognition, generative reasoning, and low-latency voice synthesis across 10 Indian regional languages and Hinglish.

2. System Architecture & Core Stack

Frontend App	Next.js 14 (App Router) + Tailwind + Framer	Modern reactive UI, Dark/Light glass aesthetic, Recharts analytics
Backend API	NestJS 10 (TypeScript Strict)	Modular microservice architecture, Guards, JWT auth, Swagger docs
Primary Database	PostgreSQL 18 + Prisma ORM	ACID relational store, Multi-tenant schema, Leads, Calls, Agents
In-Memory Store	Redis (port 6379)	BullMQ asynchronous task queues, SIP session caching, rate limits
Realtime Gateway	Socket.io (Namespace: /calls)	Live call status broadcasts, real-time waveform bars, captions
Telephony	Twilio Voice & Exotel SIP	Twiml Media Stream bridging, bidirectional raw audio websockets
Billing Engine	Razorpay + HMAC Webhooks	Automated subscription tiers, INR billing, monthly minute usage tracking

3. Dashboard User Interfaces Implemented

/dashboard/overview	COMPLETED	Operational KPIs, Recharts call volume chart, agent leaderboard, active ticker
/dashboard/agents	COMPLETED	5-step no-code builder wizard, PostgreSQL database deployment, status filter
/dashboard/calls	COMPLETED	Socket.io real-time event updates, audio waveform player, call disposition filters
/dashboard/crm	COMPLETED	10 live PostgreSQL leads, pipeline stages (Qualified, Won, Lost), lead scores
/dashboard/voices	COMPLETED	6 ElevenLabs voice profiles, pitch/speed slider controls, speech synthesis audio test
/dashboard/automations	COMPLETED	Trigger-action workflow rules (WhatsApp, SMS, Email, Webhook), template editor
/dashboard/calendar	COMPLETED	AI scheduled demo calendar, Google/Outlook sync indicator, one-click dialing
/dashboard/billing	COMPLETED	Monthly Minutes Allowance progress bar (1,847/5,000m), Razorpay checkout modal
/dashboard/analytics	COMPLETED	Multi-axis radar charts, sentiment distribution analysis, high-intent discovery
/dashboard/settings	COMPLETED	Twilio SIP credentials, API keys, timezone, 2FA policy, workspace branding
/dashboard/workspace	COMPLETED	Team seats directory, RBAC roles (Admin, Manager, Agent), colleague invite modal

4. High-Scale Engineering Blueprint (100k+ Users)

To reliably support 100,000+ active business tenants and hundreds of thousands of concurrent real-time voice calls, the following infrastructure roadmap is architected:

1. SIP Trunk & Voice Engine Clustering	Deploy self-hosted LiveKit / FreeSWITCH media clusters on AWS EKS with autoscaling. Terminate audio via WebRTC/SIP RTP packets directly into regional edge nodes across Mumbai, Singapore, and Frankfurt to maintain <30ms network jitter.
2. Sub-800ms AI Speech Pipeline	Utilize Deepgram Nova-2 streaming STT with WebSockets, stream tokens from OpenAI GPT-4o Realtime mini, and pipe output chunks to Cartesia Sonic or ElevenLabs Flash TTS without waiting for full sentences.
3. Database Horizontal Partitioning	Implement tenant-level schema isolation and read replicas via AWS Aurora Serverless v2 PostgreSQL. Route intensive reporting queries to read replicas and cache hot tenant configs in Redis Cluster.
4. Distributed Queue Concurrency	Segregate BullMQ job queues into dedicated worker pods: Outbound Call Dispatcher, Transcript Diarization Worker, Post-Call WhatsApp Dispatcher, and Razorpay Invoicing Worker.
5. Multi-Tenant Enterprise Security	Enforce AES-256 GCM encryption at rest for SIP audio recordings and DB credentials. Audit logging for GDPR, SOC-2, and Indian DPDP compliance.

5. PostgreSQL Relational Data Models (Prisma)

Tenant	id, name, slug, plan (starter/growth/business), planExpiresAt, isActive, users[], agents[]
User	id, email, passwordHash, role (super_admin, company_admin, manager, agent), tenantId
AI Agent	id, name, role, language, voiceId, businessGoal, openingScript, qualificationRules, status
Lead	id, name, phone, email, company, status (new, qualified, closed_won, closed_lost), score, notes
Call	id, direction (inbound/outbound), status, duration, recordingUrl, sentimentScore, notes, transcript
AutomationLog	id, type (whatsapp, sms, email, webhook), template, message, status (queued, sent, failed)